

VVSI Project - LSTM for stock returns

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Abstract

Forecasting equity prices and returns is a challenging task. Numerous statistical and machine-learning methods have been designed and tested for this purpose. More recently, deep neural networks (DNNs) have shown promising results in predicting equity returns. This paper investigates whether this also holds for a global Long Short-Term Memory (LSTM) model that predicts overnight returns across 1,834 tickers. Using data for NYSE and NASDAQ tickers obtained from Yahoo Finance, we define a baseline LSTM model and train it with three window sizes: 10, 30, and 100 trading days. Performance is evaluated primarily using MSE and R^2 , with RMSE, MAE, and sMAPE as complementary metrics. The results show that R^2 is concentrated around zero and that performance is weaker on the test split than on the training and validation splits. The test period includes the COVID-19 market disruption, whose increased volatility may have affected these metrics. We therefore conclude that our LSTM is not a strong generalizer across market periods and assets when returns are concentrated around zero. Its predictions favour the mean rather than asset-specific variation.

1 Introduction

1.1 Context

One of the greatest challenges in finance is accurately predicting the future price of a financial asset, whether it is an equity, option, cryptocurrency, commodity, or another market product. The reason is simple: knowing the next day's price would allow an investor to optimize potential profit. The basic idea of price forecasting is to estimate an asset's future price—one day, five days, or another period ahead—from its historical time series.

Time-series analysis is complex because it involves properties such as stationarity, ergodicity, roughness, and seasonality. Statistical methods have traditionally addressed forecasting by analysing these characteristics. However, the rapid development of machine-learning (ML) methods over the past 20 years, especially deep neural networks such as LSTMs and large language models, has increased interest in applying these models to time-series analysis, with results that can equal or outperform those of statistical methods.

1.2 Motivations

In ML, data are central to building an effective model. However, data are not always examined in the same way as in statistics, where the focus is often on understanding their

properties and designing a forecasting model for a single object of study (a local model). ML models, instead, learn characteristics from the data during training, which can be extended across multiple subjects (a global model).

This study investigates how well a baseline LSTM predicts the overnight returns of equities when trained on data from multiple tickers.

1.3 Contributions

In this project, we have two main contributions:

- A reproducible pipeline for collecting and preparing 20 years of historical data across 1,834 equities.
- A global LSTM model that predicts next-day overnight returns across many companies.

1.4 Related Work

Traditional econometric models such as SARIMA [1] have been extensively used to predict equity prices because of their intrinsic ability to account for data-specific properties such as seasonality, stationarity, and mean reversion. However, they lack the tools required to explain nonlinear behaviour.

LSTMs [2] have been widely used as an alternative to conventional recurrent neural networks because their architecture mitigates the vanishing-gradient problem while retaining an accurate chronological representation. Nevertheless, most implementations use local models.

1.5 Outline

1. **Section 2 - Background** introduces the financial and machine-learning concepts required to understand the project.
2. **Section 3 - Methods** describes the dataset preparation, forecasting problem, LSTM architecture, data splits, and evaluation methodology.
3. **Section 4 - Implementation** presents the implementation of the data pipeline, model-training process, checkpoint selection, and experiment tracking.
4. **Section 5 - Experimental Results** describes the experimental setting and compares the results obtained with input windows of 10, 30, and 100 trading days.
5. **Section 6 - Conclusions** summarizes the findings, discusses the limitations of the project, and proposes possible future developments.

2 Background

2.1 Financial Concepts

Financial market data are commonly represented through OHLCV observations. For each asset and trading day, *Open* is the price at the beginning of the trading session, *High* and

Low are respectively the highest and lowest prices observed during the session, and *Close* is the price at the end of the session. *Volume* is the number of units traded during that period. Together, these variables summarize the daily price range, its endpoints, and the level of trading activity.

Prices from different assets are not directly comparable because their levels and units differ. A return instead measures the relative change in price. The simple close-to-close return for day t is

$$r_t = \frac{C_t - C_{t-1}}{C_{t-1}}, \quad (1)$$

where C_t is the closing price on day t . A positive return represents a price increase and a negative return represents a price decrease. Since returns are relative quantities, they make movements across assets more comparable than raw prices.

This project predicts the overnight return, defined as

$$r_t^{\text{overnight}} = \frac{O_t - C_{t-1}}{C_{t-1}}, \quad (2)$$

where O_t is the opening price on day t . It measures the price movement between the previous market close and the next market open. This interval can reflect information released while the regular trading session is closed, including company announcements, macroeconomic news, and developments in other markets.

2.2 Long Short-Term Memory Networks

A Long Short-Term Memory (LSTM) network is a recurrent neural network designed to process ordered sequences while retaining information over time. At time t , an LSTM receives the current input vector x_t , the previous hidden state h_{t-1} , and the previous cell state c_{t-1} . Its behaviour is controlled by gates whose values lie between zero and one.

The forget gate determines how much of the previous cell state is retained:

$$f_t = \sigma(W_f[x_t, h_{t-1}] + b_f). \quad (3)$$

The input gate determines how much new information is written, while the candidate state contains the new information that may be stored:

$$i_t = \sigma(W_i[x_t, h_{t-1}] + b_i), \quad (4)$$

$$\tilde{c}_t = \tanh(W_c[x_t, h_{t-1}] + b_c). \quad (5)$$

The cell state is then updated by combining retained memory with new information:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (6)$$

where $\odot$ denotes element-wise multiplication. Finally, the output gate selects which part of the cell state becomes the hidden state passed to the next time step:

$$o_t = \sigma(W_o[x_t, h_{t-1}] + b_o), \quad (7)$$

$$h_t = o_t \odot \tanh(c_t). \quad (8)$$

The main contribution of the LSTM architecture is its gated cell state. Its additive update provides a more direct path through time for information and gradients, reducing the vanishing-gradient problem found in conventional recurrent neural networks. The model can therefore learn when to retain, replace, or expose information and can represent dependencies over longer periods. These properties make LSTMs suitable for financial time series, where the relevance of past observations can vary over time. LSTMs do not eliminate all optimization difficulties, but they make long-term dependencies substantially easier to learn [2].

2.3 Evaluation Metrics

Let y_i be the observed return, $\hat{y}_i$ the predicted return, $\bar{y}$ the mean observed return, and n the number of predictions. This project reports five complementary evaluation metrics.

The Mean Squared Error (MSE) is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (9)$$

MSE is the optimization loss used to train the model. Squaring the errors penalizes large mistakes more strongly than small ones. Lower values are better, although the result is expressed in squared-return units.

The Root Mean Squared Error (RMSE) is

$$\text{RMSE} = \sqrt{\text{MSE}}. \quad (10)$$

RMSE retains the stronger penalty for large errors but is expressed in the same units as the predicted return, making it easier to interpret than MSE.

The Mean Absolute Error (MAE) is

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (11)$$

MAE represents the average absolute prediction error in return units. It gives each error a linear weight and is therefore less sensitive to isolated large errors than MSE and RMSE.

The symmetric Mean Absolute Percentage Error (sMAPE) is

$$\text{sMAPE} = \frac{200}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i| + \varepsilon}, \quad (12)$$

where $\varepsilon = 10^{-8}$ prevents division by zero. sMAPE expresses error as a percentage and is bounded between 0% and 200%, with lower values being better. It is used instead of MAPE because financial returns frequently equal or approach zero and can be negative; dividing only by the observed value can therefore produce extremely large or undefined MAPE values. Despite its symmetry, sMAPE can still assign a large percentage error when both returns are close to zero, so it should be interpreted together with the absolute-error metrics.

The coefficient of determination (R^2) is

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (13)$$

An R^2 of 1 represents perfect predictions, while 0 indicates performance equivalent to always predicting the mean observed return. A negative value means that the model performs worse than that mean baseline. Unlike the four error metrics, a higher R^2 is better.

3 Methods

3.1 Data

Data processing was a central part of the pipeline.

The **first step** collected 5,984 NASDAQ and NYSE ticker symbols. Their different share-class representations were filtered, deduplicated, and normalized to the hyphenated format expected by Yahoo Finance.

The **second step** downloaded the maximum available daily OHLCV history from Yahoo Finance through the `yfinance` package. A rate-aware scheduler checkpointed each request and retried rate limits and transient failures, producing usable histories for 5,842 tickers. Automatic price adjustment made historical OHLC prices consistent after splits and dividends, avoiding artificial jumps in the return series.

The **third step** selected the 1,834 tickers with complete, non-missing OHLCV coverage from January 1, 2006 to January 1, 2026. This universe contains 9,226,853 daily observations before constructing returns and rolling windows.

The **fourth and final step** derived the relative daily range $(H_t - L_t)/C_t$, log-volume $\ln(V_t)$, overnight return $(O_t - C_{t-1})/C_{t-1}$, and intraday return $(C_t - O_t)/O_t$. Log-volume was standardized per ticker using training-only statistics, and the next-day overnight return was used as the target. The chronological split uses 2006–2015 for training, 2016–2018 for validation, and 2019–2025 for testing.

3.2 Model

The model uses a custom PyTorch LSTM cell instead of the built-in `nn.LSTM`, providing explicit control over its gates and initialization. It has four input features, one recurrent layer with a 64-dimensional hidden and cell state, and a linear output layer that predicts one overnight-return value from the final hidden state. Gate and output weights use Xavier uniform initialization, while the forget-gate bias is initialized to one to encourage the retention of earlier information at the start of training. Three otherwise identical models use window sizes of 10, 30, and 100 trading days to evaluate the effect of different historical horizons. We use Adam as our optimizer and MSE as our objective loss.

4 Implementation

The system trains the global LSTM on the aggregated multi-ticker dataset. At each epoch, the training and validation metrics are logged to Weights & Biases (W&B), allowing their evolution to be monitored throughout training. Whenever the validation MSE improves, the current checkpoint replaces the previous one, so only the best model is retained. After training, this checkpoint is reloaded and evaluated on the training, validation, and test sets. The final metrics are presented in a summary table, together with the 15 test tickers with the lowest MSE and highest R^2 for a per-ticker comparison.

5 Experimental Results

5.1 Goals

The main goal of the experiment is as follows:

- Verify the accuracy of a global LSTM when predicting tickers' overnight returns.

5.2 Setting

To provide an accurate and comprehensive evaluation, we trained the model using three window sizes to assess the effect of historical context on its predictions. Each model was trained for 50 epochs with a batch size of 128, a hidden size of 64 and a learning rate of 10^{-4} .

5.3 Technical Evaluation

The overnight returns of the 15 sampled tickers fluctuate around zero, with occasional positive and negative spikes in every split (Figure 1). This behaviour makes small numerical errors possible even when a model explains little of the variation between individual observations.

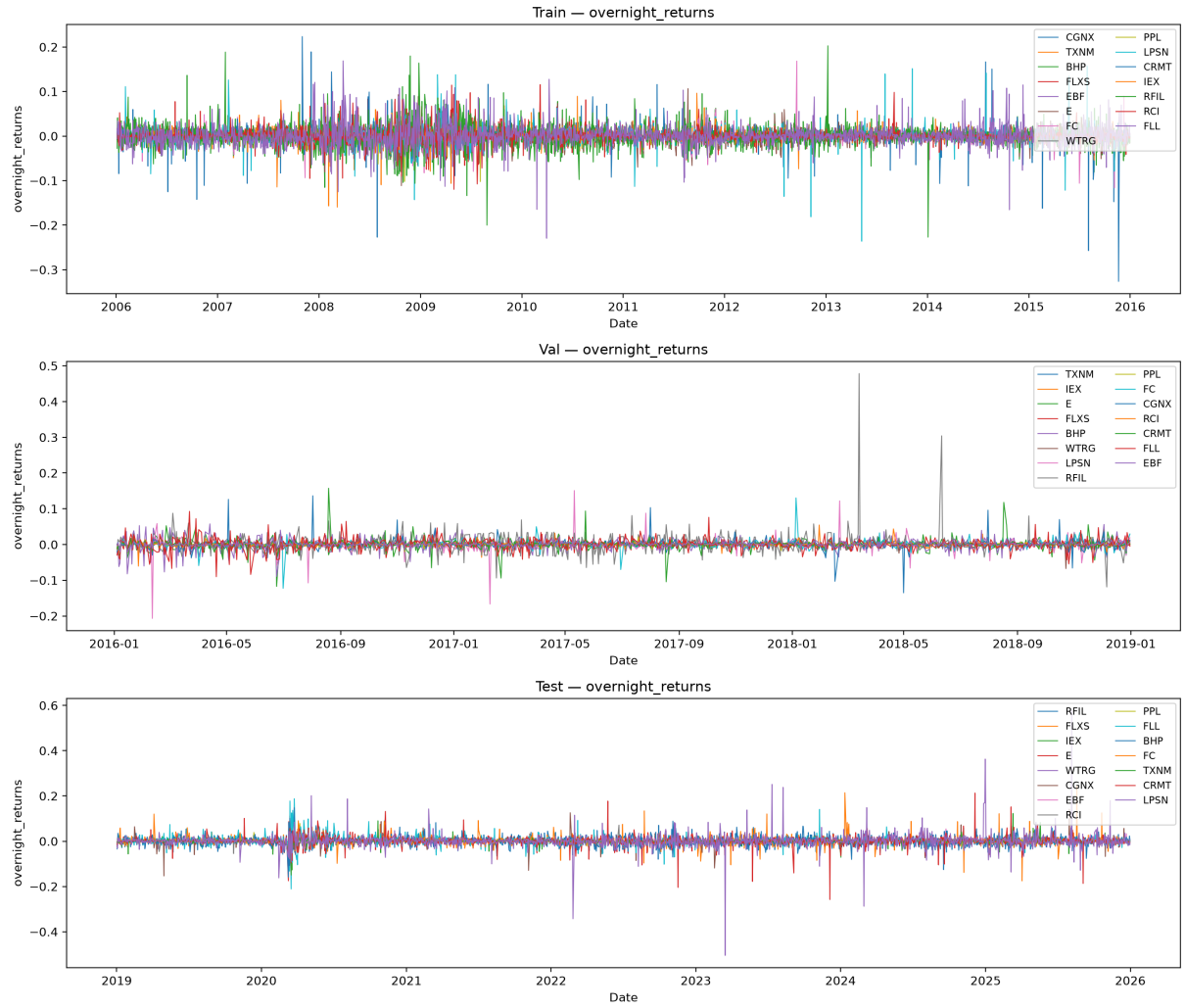


Figure 1: Overnight returns for a deterministic sample of 15 tickers in the training, validation, and test periods.

Figures 2–4 show how training and validation MSE evolve over the 50 epochs. Training MSE decreases steadily in all three runs, whereas validation MSE remains nearly constant with only small fluctuations. This difference explains why the final evaluation reloads the checkpoint with the lowest validation MSE rather than automatically using the last epoch.

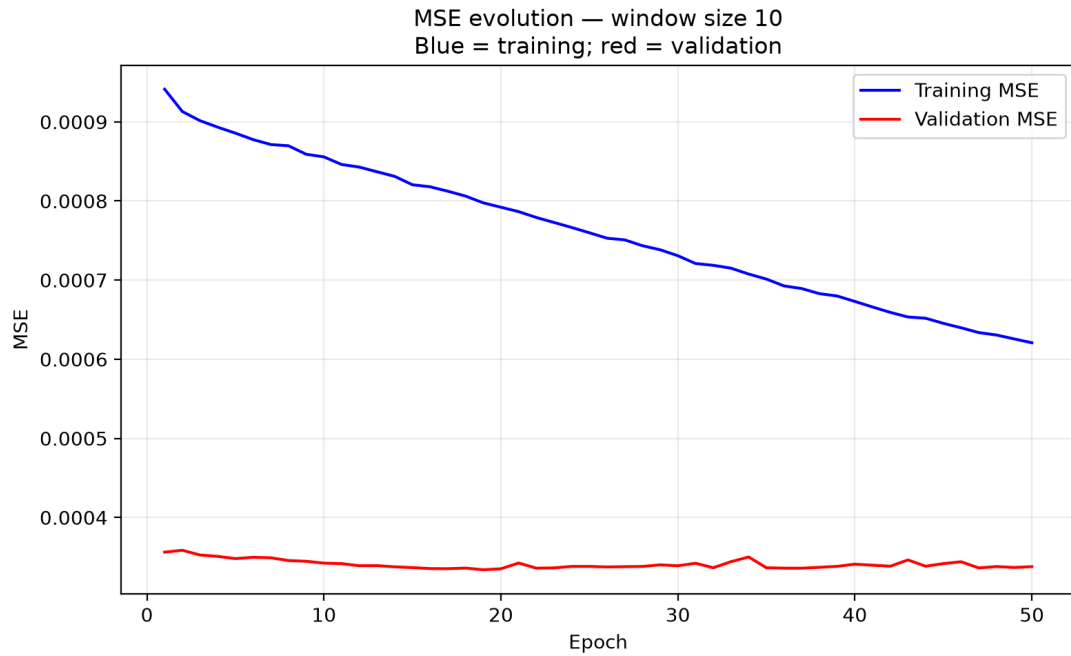


Figure 2: MSE evolution for window size 10. Blue denotes training MSE and red validation MSE.

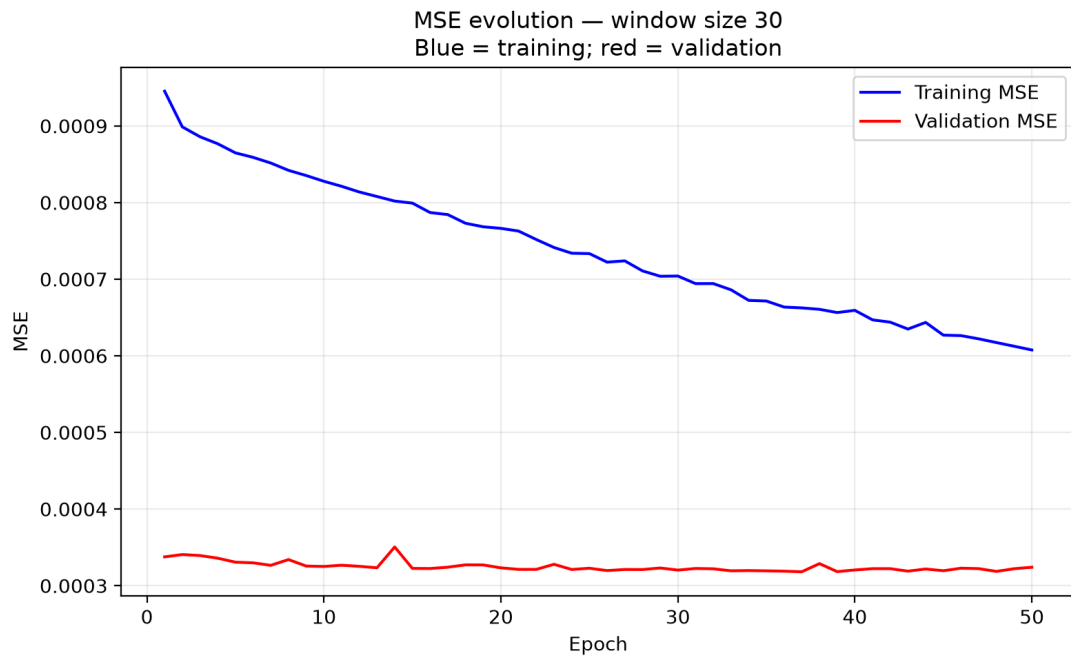


Figure 3: MSE evolution for window size 30. Blue denotes training MSE and red validation MSE.

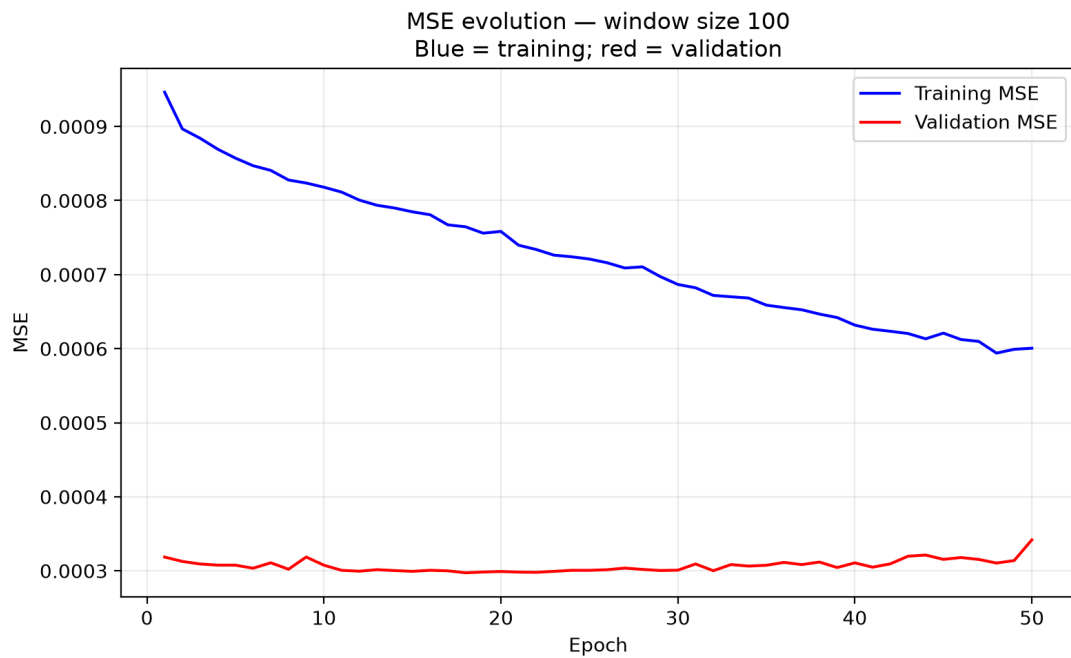


Figure 4: MSE evolution for window size 100. Blue denotes training MSE and red validation MSE.

For R^2 , removing the lowest and highest 0.5% of ticker-level values makes the central 99% easier to inspect (Figures 5–7). This view is used because tickers with very low return variance can produce extreme negative R^2 values that compress the central distribution. For every window size, most values are concentrated near zero and the distribution is left-skewed. The negative tail is still pronounced for window size 100, even after trimming the extremes. Therefore, increasing the historical window did not produce a material improvement in out-of-sample explanatory power.

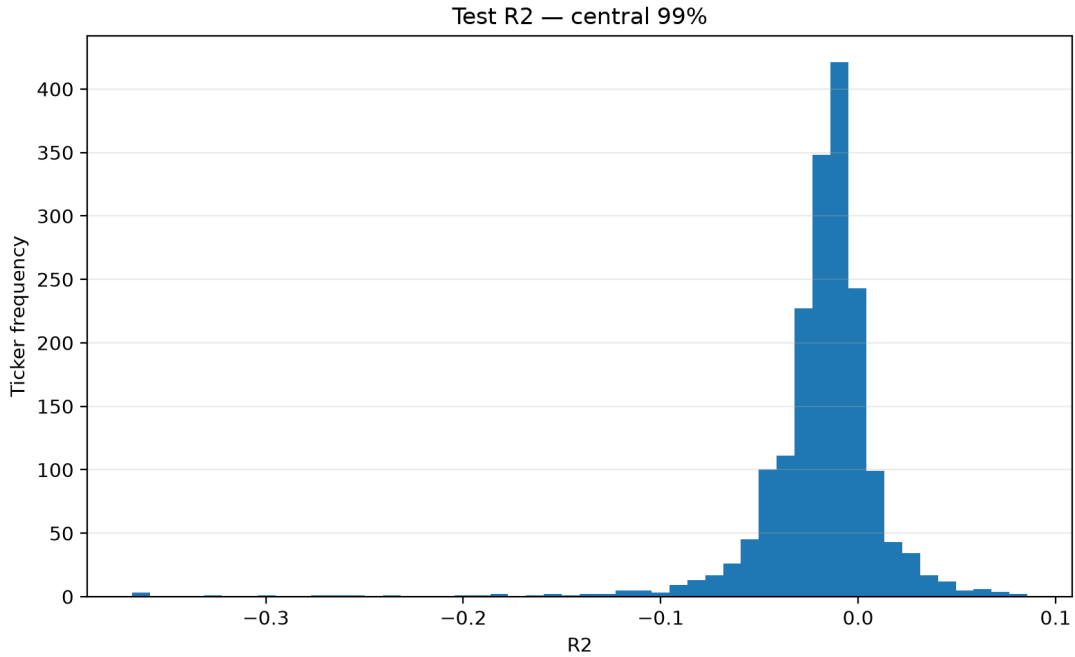


Figure 5: Central 99% of the test R^2 distribution for window size 10.

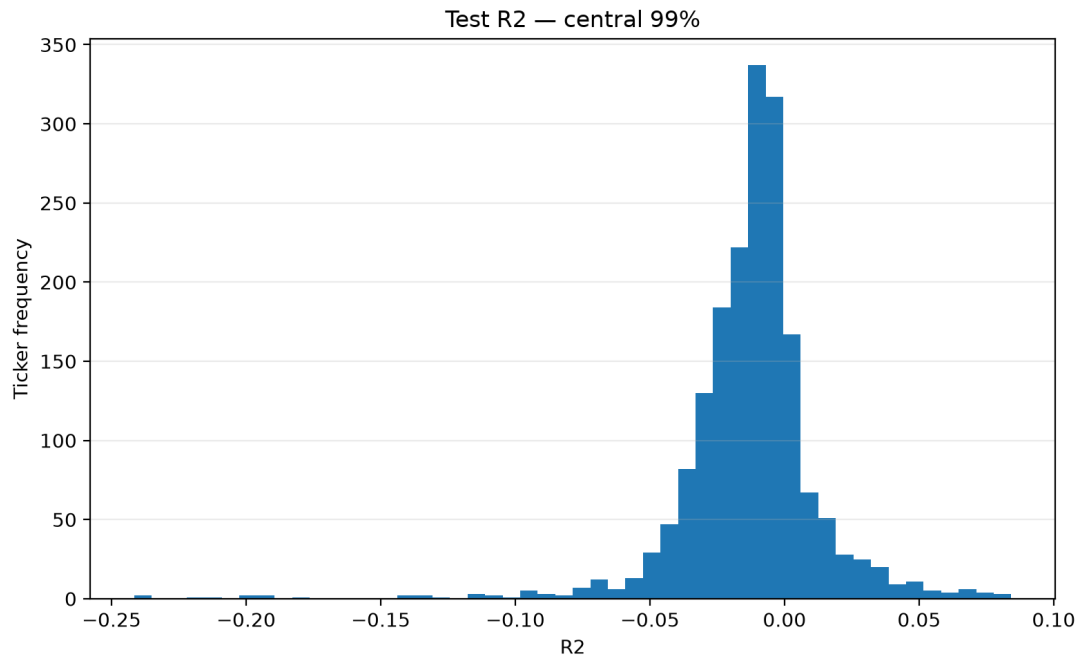


Figure 6: Central 99% of the test R^2 distribution for window size 30.

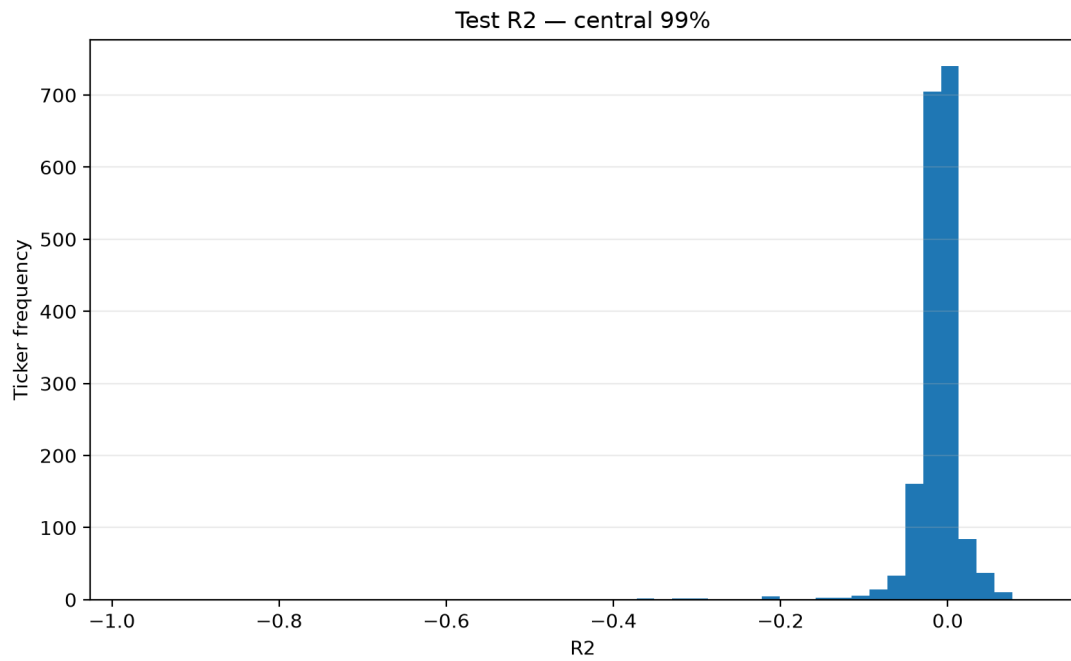


Figure 7: Central 99% of the test R^2 distribution for window size 100.

The sets of 15 test tickers with the lowest MSE are broadly similar across window sizes (Figures 8–10), although their ordering and best values change. The ticker with the lowest MSE does not necessarily have the largest R^2 , because MSE measures error size while R^2 measures improvement over the ticker’s mean-return baseline. Overnight returns are generally close to zero. Consequently, a small absolute prediction error can produce a large sMAPE. Therefore, sMAPE should be interpreted together with MSE, RMSE, and MAE.

Test best 15 tickers by MSE
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
LEO	1750	2.20975e-05	0.0047008	0.00312467	156.664	-0.0791603
TVE	1750	2.44761e-05	0.00494733	0.0035458	150.686	-0.274117
TVC	1750	2.99112e-05	0.00546912	0.00374319	152.057	-0.367683
CHSCP	1750	3.06188e-05	0.00553342	0.00382735	152.734	0.0103414
NXP	1750	3.15718e-05	0.00561888	0.00318338	157.842	-0.0468658
CHT	1750	3.23304e-05	0.00568598	0.00397539	143.94	-0.0369333
WEC	1750	3.98184e-05	0.00631019	0.00394314	149.239	-0.030379
BCE	1750	4.05673e-05	0.00636925	0.00388336	153.728	-0.024247
WCN	1750	4.0724e-05	0.00638154	0.00394905	148.478	-0.0387762
ED	1750	4.27216e-05	0.00653617	0.00421986	153.384	-0.0581402
LNT	1750	4.30172e-05	0.00655875	0.00417822	154.32	-0.0473914
DUK	1750	4.41662e-05	0.00664576	0.00398808	148.302	-0.0215203
AEE	1750	4.43493e-05	0.00665953	0.0040787	150	-0.031964
VLGEA	1750	4.49792e-05	0.00670665	0.00461672	160.901	-0.0182675
CMS	1750	4.52734e-05	0.00672855	0.00411067	147.282	-0.0440651

Figure 8: Best 15 test tickers by MSE for window size 10. Red denotes the best metric value and blue the second best; larger R^2 is better.

Test best 15 tickers by MSE
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
TVE	1730	2.1081e-05	0.00459141	0.00328	150.815	-0.110502
TVC	1730	2.14437e-05	0.00463074	0.00329371	147.265	-0.0957387
LEO	1730	2.26664e-05	0.00476093	0.00317553	173.003	-0.0991045
CHSCP	1730	3.08987e-05	0.00555866	0.00377341	163.013	0.00229623
CHT	1730	3.15501e-05	0.00561695	0.00394149	166.78	-0.00417721
NXP	1730	3.48201e-05	0.00590086	0.00348851	161.373	-0.142206
BCE	1730	4.01305e-05	0.00633486	0.00386283	162.423	-0.00528666
WEC	1730	4.04837e-05	0.00636268	0.00398836	164.594	-0.0381323
WCN	1730	4.15066e-05	0.00644256	0.00404104	163.878	-0.049549
ED	1730	4.2303e-05	0.00650407	0.00418576	161.899	-0.0389086
LNT	1730	4.28032e-05	0.00654242	0.00416864	165.306	-0.0345426
DUK	1730	4.45778e-05	0.00667666	0.00402085	161.85	-0.0222879
VLGEA	1730	4.48403e-05	0.00669629	0.0045875	167.278	-0.00917993
AEE	1730	4.49542e-05	0.00670479	0.00415651	163.765	-0.036442
ATO	1730	4.54704e-05	0.00674317	0.00407618	161.985	-0.0147917

Figure 9: Best 15 test tickers by MSE for window size 30. Red denotes the best metric value and blue the second best; larger R^2 is better.

Test best 15 tickers by MSE
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
LEO	1660	2.25924e-05	0.00475314	0.00314825	151.523	-0.0668777
TVE	1660	2.30474e-05	0.00480077	0.00349545	144.179	-0.201426
TVC	1660	2.35616e-05	0.00485403	0.00351626	144.484	-0.216245
CHSCP	1660	3.27218e-05	0.0057203	0.00388517	147.293	-0.0322129
NXP	1660	3.27677e-05	0.00572431	0.00333972	151.138	-0.0358197
CHT	1660	3.3073e-05	0.00575091	0.00401381	145.104	-0.0309008
WEC	1660	4.09464e-05	0.00639893	0.00402338	150.078	-0.0141531
WCN	1660	4.14809e-05	0.00644057	0.00396551	147.05	-0.0373545
BCE	1660	4.22959e-05	0.00650353	0.00398905	153.349	-0.0266893
ED	1660	4.33045e-05	0.00658062	0.00423869	154.823	-0.0287538
LNT	1660	4.37816e-05	0.00661676	0.00422484	154.841	-0.0253115
VLGEA	1660	4.5264e-05	0.00672785	0.00463014	160.468	-0.0114063
DUK	1660	4.56072e-05	0.00675331	0.00407705	150.492	-0.0120658
AEE	1660	4.57344e-05	0.00676272	0.00415624	149.679	-0.0230066
CMS	1660	4.65854e-05	0.00682535	0.00417556	149.584	-0.0278109

Figure 10: Best 15 test tickers by MSE for window size 100. Red denotes the best metric value and blue the second best; larger R^2 is better.

Figures 11–13 instead rank the test tickers by descending R^2 . This ranking identifies the tickers for which the model improves most over the ticker-specific mean-return baseline, rather than those with the smallest absolute errors. MFA obtains the highest R^2 for all three window sizes, while the maximum falls from 0.574 for window size 10 to 0.545 for window size 100. Comparing these tables shows that lower MSE does not necessarily correspond to higher R^2 .

Window size 10 — Test best 15 tickers by R^2
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
MFA	1750	0.000627839	0.0250567	0.00845721	160.202	0.57446
CNTY	1750	0.000656252	0.0256174	0.0119825	166.473	0.114501
SCWO	1470	0.00721698	0.0849528	0.0400698	171.997	0.112625
BVFL	1135	0.000586119	0.0242099	0.0134731	170.309	0.112302
OMEX	1750	0.00116263	0.0340973	0.0176782	164.874	0.107905
TSSI	1566	0.00214412	0.0463046	0.029902	167.764	0.100366
LINK	1625	0.00154136	0.0392602	0.0240544	164.771	0.0936709
ACFN	1646	0.00240348	0.0490253	0.0331507	169.382	0.0936043
ADAM	1750	0.000503294	0.0224342	0.00835371	163.712	0.0921112
BANFP	1121	0.000215343	0.0146746	0.00868898	159.561	0.0857399
PYT	886	0.000242016	0.0155569	0.00981588	153.427	0.0787267
SMTI	1725	0.00246959	0.0496949	0.0223298	170.727	0.0752285
NXPL	1295	0.00992334	0.099616	0.0419874	170.566	0.0740138
DSWL	1729	0.000372629	0.0193036	0.0131084	161.207	0.0737127
DGICB	1014	0.00130038	0.0360608	0.0243876	169.307	0.0678687

Figure 11: Best 15 test tickers by R^2 for window size 10. Red denotes the best metric value and blue the second best; larger R^2 is better.

Window size 30 — Test best 15 tickers by R^2
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
MFA	1730	0.000662164	0.0257325	0.00838382	175.33	0.556299
FNRN	1247	0.000178661	0.0133664	0.00877195	150.079	0.165865
TSSI	1546	0.00202908	0.0450453	0.0291036	163.288	0.150266
EFSL	1138	0.000251057	0.0158448	0.0097388	156.02	0.149524
OMEX	1730	0.00112282	0.0335085	0.0175578	169.582	0.142173
LINK	1605	0.0014417	0.0379697	0.0233842	160.721	0.107951
ACFN	1626	0.00233489	0.0483207	0.032589	169.37	0.101293
BVFL	1115	0.00059265	0.0243444	0.0134442	174.021	0.0964152
SMID	1718	0.000452897	0.0212814	0.0141557	163.987	0.0843864
SOTK	1610	0.000945143	0.0307432	0.019891	163.1	0.0842451
CAPS	386	0.0221873	0.148954	0.0644292	176.448	0.0784319
GL	1730	0.000146972	0.0121232	0.00686929	166.621	0.0781869
NTWK	1729	0.000521682	0.0228404	0.015122	167.85	0.0757045
ADAM	1730	0.000518445	0.0227694	0.00841169	177.458	0.0753702
GME	1730	0.00434155	0.0658904	0.0209643	159.149	0.0750476

Figure 12: Best 15 test tickers by R^2 for window size 30. Red denotes the best metric value and blue the second best; larger R^2 is better.

Window size 100 — Test best 15 tickers by R2
Red = best; blue = second best

ticker	observations	mse	rmse	mae	smape	r2
MFA	1660	0.000707435	0.0265976	0.00874336	160.019	0.545011
EFSL	1068	0.000255154	0.0159735	0.00995342	145.856	0.163209
OMEX	1660	0.00113028	0.0336197	0.0175924	168.724	0.160791
TSSI	1476	0.00207834	0.0455888	0.0297729	161.264	0.154677
BANFP	1031	0.000221847	0.0148945	0.00904757	150.28	0.125565
FNRN	1177	0.000195976	0.0139991	0.00950534	135.711	0.115313
LINK	1535	0.0013506	0.0367506	0.0228717	161.385	0.108758
ACFN	1556	0.00228092	0.0477589	0.0324985	162.454	0.108021
BVFL	1045	0.000553749	0.0235319	0.0130138	169.087	0.105411
CNTY	1660	0.000700336	0.0264639	0.0122557	171.33	0.0990471
ADAM	1660	0.000537336	0.0231805	0.00868347	172.812	0.0775593
SMID	1648	0.00047157	0.0217157	0.0146245	163.986	0.0769412
DGICB	924	0.00125683	0.0354518	0.0246093	167.421	0.0731747
SCWO	1380	0.00501507	0.0708172	0.0351214	173.996	0.0730322
NTWK	1659	0.000539019	0.0232168	0.0154443	167.522	0.0728231

Figure 13: Best 15 test tickers by R^2 for window size 100. Red denotes the best metric value and blue the second best; larger R^2 is better.

Figure 14 compares the return distributions of the 15 test tickers with the highest R^2 for window size 10 across the three data splits. Although all 15 distributions are strongly concentrated around zero, MFA, the ticker with the highest R^2 , has the narrowest observed return distribution, while the remaining tickers have wider distributions.

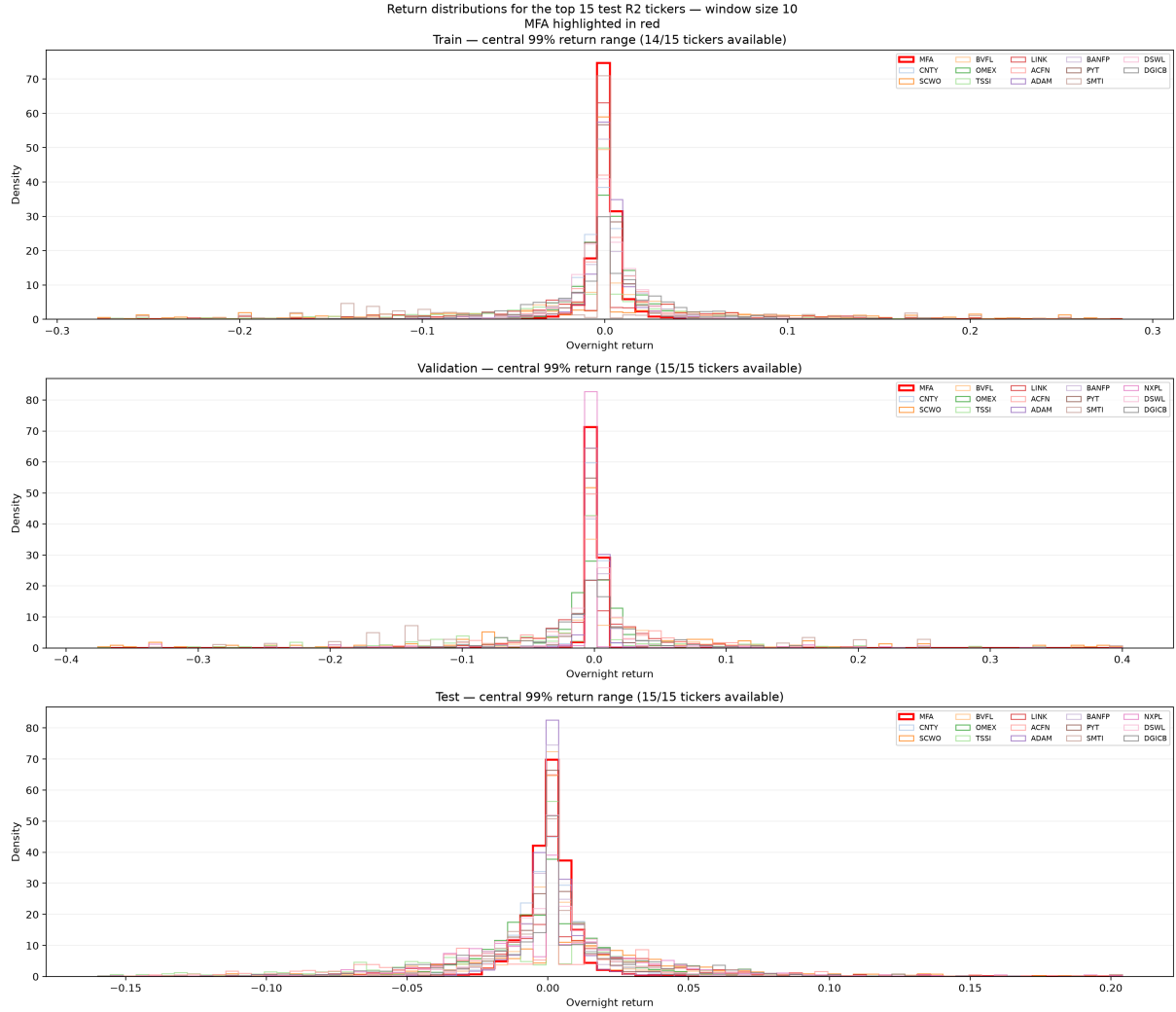


Figure 14: Return distributions across the training, validation, and test splits for the 15 test tickers with the highest R^2 at window size 10.

Tables 1–3 compare the aggregate metrics across training, validation, and test data. All three models obtain their lowest MSE, RMSE, and MAE on validation data, while their test R^2 values remain close to zero. The test split covers 2019–2025 and therefore includes the COVID-19 market disruption, a period of elevated volatility that may have contributed to the weaker test metrics. Nevertheless, the results indicate that the model does not generalize to the test period as effectively as it fits the training data. Window size 10 provides the lowest test MSE and the largest test R^2 , but the differences between window sizes are small.

Table 1: Final metrics from the best-validation LSTM checkpoint for window size 10. Red denotes the best value and blue the second best.

split	observations	tickers	mse	rmse	mae	smape	r2
train	4431943	1831	0.000786084	0.0280372	0.0101335	159.313	0.179536
val	1336174	1833	0.000333872	0.0182722	0.0076154	158.063	0.0587641
test	3183324	1834	0.00292529	0.0540859	0.0100078	160.94	0.0127316

Table 2: Final metrics from the best-validation LSTM checkpoint for window size 30. Red denotes the best value and blue the second best.

split	observations	tickers	mse	rmse	mae	smape	r2
train	4395323	1831	0.000647808	0.0254521	0.0099813	165.242	0.313234
val	1299514	1833	0.000318126	0.0178361	0.00735804	165.735	0.06136
test	3146644	1834	0.00295422	0.0543527	0.0100009	168.317	0.0123785

Table 3: Final metrics from the best-validation LSTM checkpoint for window size 100. Red denotes the best value and blue the second best.

split	observations	tickers	mse	rmse	mae	smape	r2
train	4267316	1827	0.000748387	0.0273567	0.0100926	161.461	0.206554
val	1171276	1831	0.000297358	0.0172441	0.00718821	159.171	0.0335147
test	3018264	1834	0.0029948	0.0547248	0.0101422	164.621	0.0101495

Table 4 summarizes the three experiments by taking the arithmetic mean of each split-level metric across the window sizes.

Table 4: Average final metrics across window sizes 10, 30, and 100. Each value is the arithmetic mean of the corresponding split metric from the three best-validation checkpoints. Red denotes the best value and blue the second best.

split	mse	rmse	mae	smape	r2
train	0.000727426	0.0269487	0.0100691	162.005	0.233108
val	0.000316452	0.0177841	0.00738722	160.99	0.0512129
test	0.0029581	0.0543878	0.0100503	164.626	0.0117532

6 Conclusions

6.1 Results

We conclude that our LSTM is not a strong generalizer for overnight returns, which are concentrated around zero. Its predictions remain close to the mean, possibly because MSE as the objective function penalizes large errors quadratically. Averaged across the three window sizes, R^2 falls from 0.233 on the training split to 0.051 on validation and 0.012 on testing, supporting the conclusion that performance deteriorates out of sample. Among the 15 test tickers with the highest R^2 , MFA achieves the largest value for every window size. The distribution pattern supports the observation that the model prioritizes predictions near the zero return mean.

6.2 Limitations

Our major limitation was computational power. Due to the lack of stronger computers, we weren't able to fully evaluate all the different combinations of hyperparameters.

6.3 Future Work

As for improvements, we could analyse daily returns, which cover a longer price interval and could exhibit different behaviour. A comprehensive evaluation of different hyperparameters would also allow relationships beyond the historical window length to be investigated and determine whether a baseline LSTM with a different configuration might be successful. Finally, using MAE as the loss function would reduce the influence of outliers on model updates, allowing the model to respond less aggressively to extreme errors.

References

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